8.2 Integration by Parts

Technique for the integral of the form

$$\int f(x)\, g(x)\, dx,$$

where one part can be **differentiated** and another part can be **integrated** (easily).

Example: $\int x \sec^2 x \, dx$

It is useful when f can be differentiated repeatedly to become zero and g can be integrated repeatedly without difficulty.

Example: $\int x^2 e^x \, dx$

Product Rule in Integral Form

Recall that $\frac{d}{dx}f(x)g(x) = f(x)g'(x) + f'(x)g(x)$.

Integrating on both sides wrt x, we get

Integration by parts formula:

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx$$

Alternative Form:

$$\int u\,dv = uv - \int v\,du$$

The key to applying integration by parts is to break the integral to a suitable u and dv.

Example Using Integration by Parts

Find $\int x \cos x\,dx$.

Let $dv = \cos x\,dx$ and $u = x$.

There are four choices for u and dv in the above example. Namely, for $\int x \cos x \, dx$, we can choose

- $dv = x \cos x dx, u = 1$
- $dv = \cos x dx, u = x$
- $dv = dx, u = x \cos x$

- $dv = x dx, u = \cos x$

Example Integral of the Natural Logarithm

Find $\int \ln x \, dx$.

Example

1. Find $\int \tan^{-1} x \, dx$.

2. Find $\int \sin^{-1} x \, dx$.

Example Repeated Use of Integration by Parts

Evaluate $\int x^2 e^x \, dx$.

In general, we can use integration by parts to evaluate $\int x^n e^x \, dx$ for any positive integer n.

(We will see this again later.)

Example Solving for the Unknown Integral

Evaluate $\int e^x \cos x \, dx$.

Example Reduction Formula

Obtain a formula that expresses the integral

$$\int \cos^n x \, dx$$

in terms of an integral of a lower power of $\cos x$.

$$\int \cos^n x\, dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} \int \cos^{n-2} x\, dx$$

$\int \cos^3 x\, dx =$

$\int \cos^4 x\, dx =$

Example: Substitution + By Parts

- $\int e^{\sqrt{2x-1}}\, dx =$

- $\int \cos(\ln x)\, dx =$

Evaluating Definite Integrals by Parts

Integration by Parts Formula for Definite Integrals

- $\int_a^b f(x)g'(x)dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x)dx$
- $\int_a^b udv = [uv]_a^b - \int_a^b vdu$

Example

Evaluate $\int_0^4 xe^{-x}\,dx$

Evaluate $\int_{2/\sqrt{3}}^{2} x\sec^{-1}x\,dx$. *Ans:* $\frac{5\pi}{9} - \frac{1}{2}(\sqrt{3} - \frac{1}{\sqrt{3}})$

Tabular Integration

For integration by parts with many repetitions.

A good way to organise the calculations.

$$\int f(x)\, g(x)\, dx$$

We want to be able to differentiate $f(x)$ repeatedly to become zero, and we want to be able to integrate $g(x)$ repeatedly (without much difficulty).

EXAMPLE 7 Evaluate

$$\int x^2 e^x \, dx.$$

Solution With $f(x) = x^2$ and $g(x) = e^x$, we list:

$f(x)$ and its derivatives		$g(x)$ and its integrals
x^2	(+)	e^x
$2x$	(−)	e^x
2	(+)	e^x
0		e^x

Example Evaluate $\int x^4 e^{-2x} dx$ using tabular integration.

Let $f(x) = x^4$ and $g(x) = e^{-2x}$.

$f(x)$ and its derivatives	$g(x)$ and its integrals
$f(x) = x^4$	$g(x) = e^{-2x}$

$\int x^4 e^{-2x} dx =$